



**TECHNISCHE
UNIVERSITÄT
DRESDEN**

Barkhausenbau, Raum BAR 60
Helmholtzstraße 18, 01062 Dresden
+49 351 463-33547
Robert.Rosenkranz@tu-dresden.de

**To
whom it may concern,**

January 15, 2025

I am pleased to write this letter of recommendation for Israt Jahan Tulin, who served as a Research Assistant at the Acoustics and Haptics Department of Technische Universität Dresden. Israt's work focused on utilizing advanced technologies such as Unreal Engine, 3D Phantom Device, Haptic Gloves, Arduino microcontroller, Oculus, and the Unreal Haptics plugin.

One of Israt's notable projects involved leveraging the Unreal Haptics plugin, developed by Universität Bremen. This plugin enables both 3-DOF and 6-DOF haptic rendering in Unreal Engine 4, integrating a state-of-the-art collision detection library for fast and stable force and torque computations. After studying the published paper on UnrealHaptics, Israt successfully implemented the plugin in our workstation at the Acoustics and Haptics Department. Her ability to adapt and configure the plugin based on the research paper was instrumental in achieving functional haptic rendering in our applications. This effort demonstrated her deep understanding of haptic technology and her capacity to apply theoretical knowledge to practical use cases.

Additionally, Israt contributed to the Vibro *FingrTac* and *KinesCeTI* projects, which utilized Leap Motion to track hand movements and provide haptic feedback through vibrations. She demonstrated her ability to work on interdisciplinary projects by collaborating with Dr.-Ing. Thomas Hulin, head of the Telemanipulation Research Group at the German Aerospace Center (DLR), and researchers from Universität Bremen to ensure system compatibility. Her contributions included refactoring C++ scripts, creating dedicated class templates for haptic devices, and integrating OSC and Pure Data for seamless data communication. She also measured latency using various sensors and ensured project compatibility with both CHAI3D and Unreal Engine environments. Furthermore, she implemented reliable vibration feedback through 3 Linear Resonant Actuators (LRAs), wearable on three fingers. These efforts significantly enhanced the functionality, versatility, and robustness of our haptic feedback systems.

Israt's technical skills, combined with her dedication and ability to work collaboratively, make her an outstanding candidate for any research or development role. She has consistently demonstrated a high level of competence and innovation, and I am confident that she will continue to excel in her future endeavors.

Sincerely,

Dr. Robert Rosenkranz